

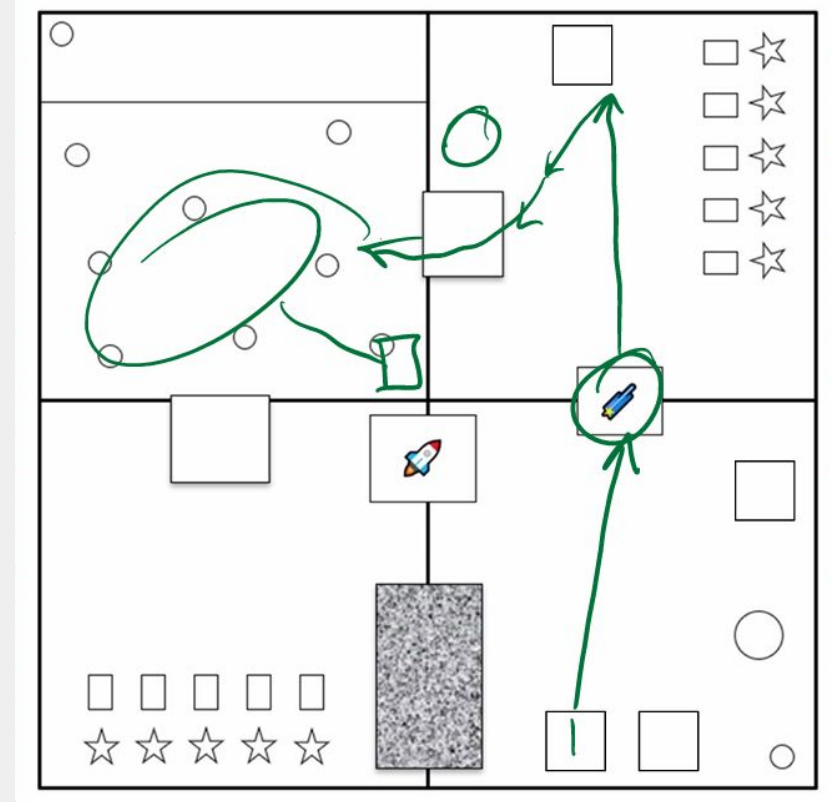
# Lightning McQueens

Team 101



# Role in Squad Strategy

- Push Asteroid
- Pick up cubes in Mars and in Space and deliver to Area 51/Rocket
- Justification:
  - RMP 1 is in the best position to push the asteroid to Q4.
  - Once RMP 1 is in Q4 it can easily grab the cubes in the same quadrant and in Mars then bring them to the Rocket/Area 51



# Functional Requirements

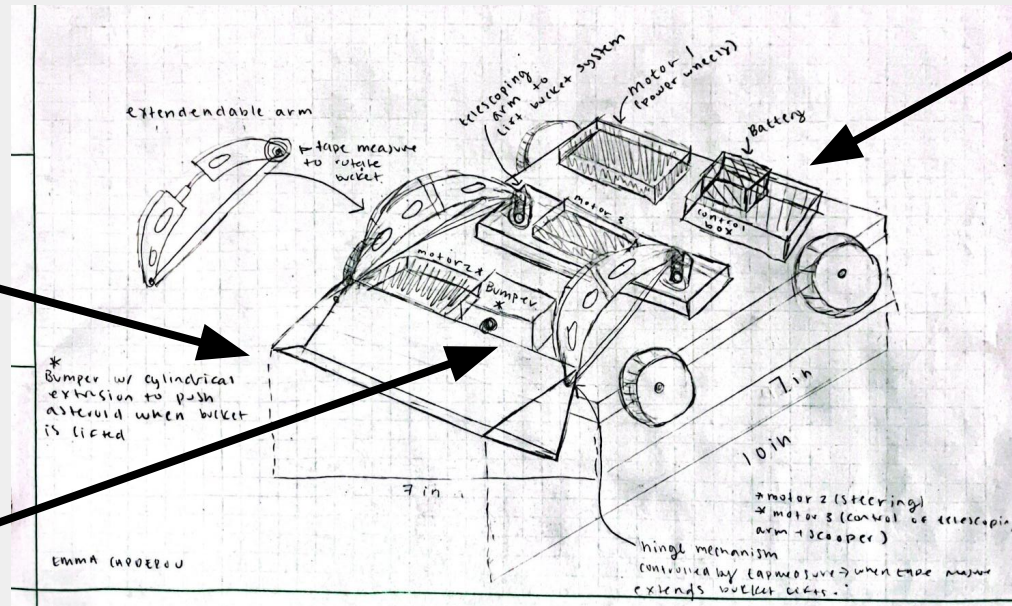
- RMP1 must be able to individually push an object weighing  $18 \text{ lbs} \pm 2 \text{ lbs}$ .
  - Designed specifically to be slightly higher than the weight of the asteroid. If this req is not met then the ability to meet additional functional requirements may increase exponentially/decrease.
- RMP1 must push the asteroid against the wall in quadrant IV within  $40 \text{ seconds} \pm 5 \text{ seconds}$ .
  - Moving the asteroid is integral to our squad's overall success, as many RMPs are planned to move directly from Q1 to Q4 but aren't planned to cross the space-time continuum. It must be done in a timely manner to ensure the other RMPs have time to complete their tasks.
- RMP1 must be able to collect  $3 \pm 1$  cubes at a time.
  - Our RMP needs to be able to collect the cubes efficiently because we are the major cube scorer in our squad. If we are unable to complete this task, our squad will be unable to score many points.



# Design Concepts Considered

Scoop to of width  
7in to collect  $3 \pm 1$   
cube

Bumper to  
individually push the  
asteroid weighing  $18$   
 $\text{lb} \pm 2$

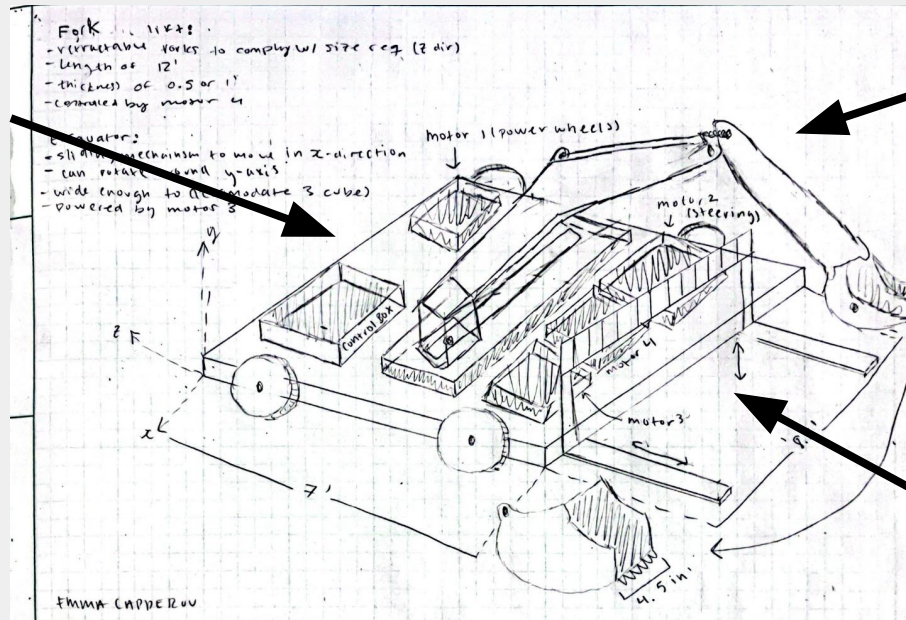


2 motors for rear  
wheels to push  
asteroid within  $40$   
 $\pm 5$  sec



# Design Concepts Considered

2 motors to push  
asteroid in  $40 \pm 2$   
sec



Scooping arm of  
width 4.5 in to  
collect  $3 \pm 1$  cubes  
at a time

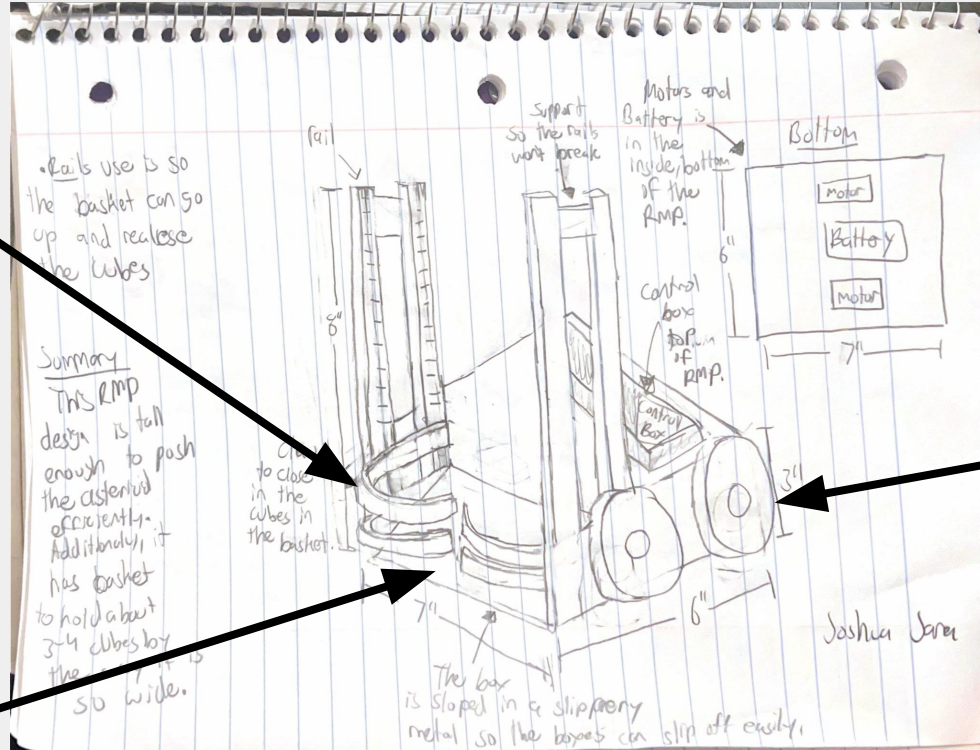
Forklift to  
individually push  
asteroid weighing  
 $18 \pm 2$  lb



# Design Concepts Considered

Able to individually push an object weighing  $18\text{lbs} \pm 2\text{lbs}$ .

Able to collect  $3 \pm 1$  cubes at a time.

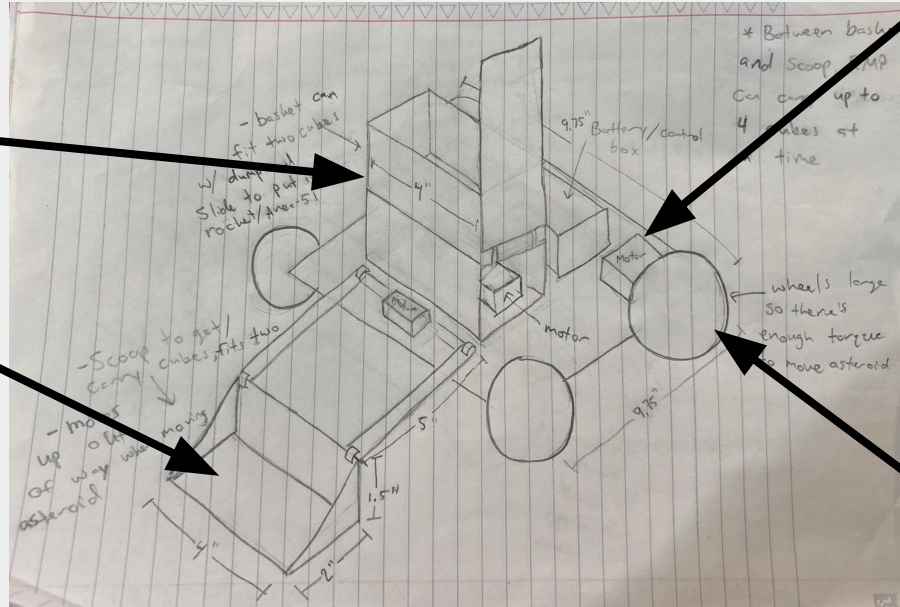


Can push the asteroid against the wall in quadrant IV within  $40\text{ seconds} \pm 5\text{ seconds}$



# Design Concepts Considered

Scoop and basket with 4" width to fit  $3 \pm 1$  cubes total.



Double gearbox motor so RMP can move asteroid in  $40 \pm 5$  seconds.

Large wheels so there is enough torque to push  $18 \pm 2$  lbs.



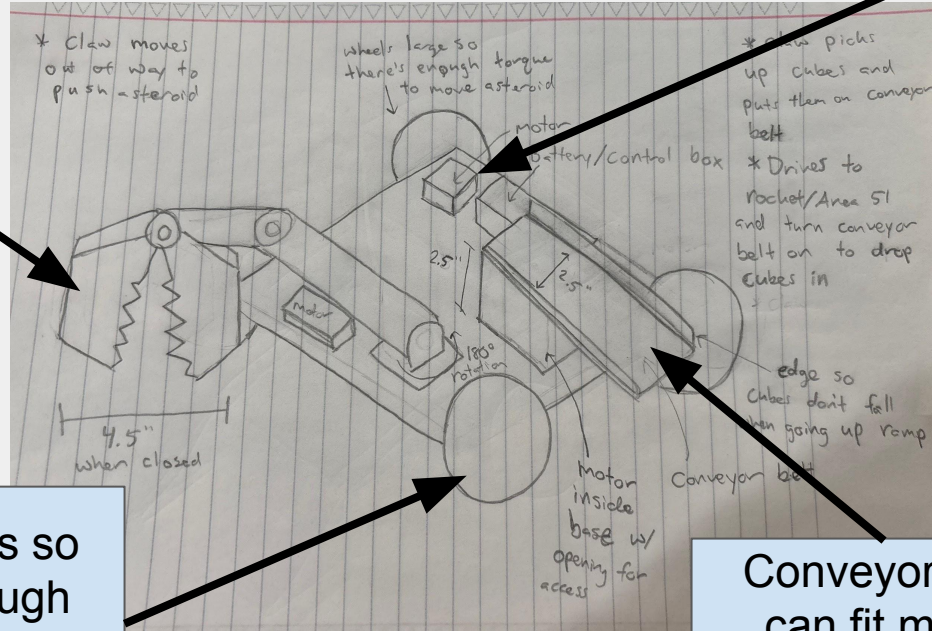
# Design Concepts Considered

Closed claw width of 4.5" to fit up to 2 cubes.

Large wheels so there is enough torque to push 18  $\pm$  2 lbs.

Conveyor belt can fit more cubes to carry 3  $\pm$  1 total.

Double gearbox motor so RMP can move asteroid in 40  $\pm$  5 seconds.



# Pugh Chart

| Requirement                                                                                     | Weight | Design concept 1 | Design concept 2 | Design concept 3 | Design concept 4 |
|-------------------------------------------------------------------------------------------------|--------|------------------|------------------|------------------|------------------|
| Must be able to individually push an object weighing <u>18lbs</u> <del>± 2lbs.</del>            | 5      | 0                | +1               | 0                | +1               |
| Must push the asteroid against the wall in quadrant IV within 40 seconds <del>± 5 seconds</del> | 5      | 0                | +1               | 0                | 0                |
| Must be able to collect 3 ± 1 cubes at a time.                                                  | 3      | 0                | 0                | +1               | +1               |
| Manufacturability                                                                               | 3      | 0                | -1               | -1               | -1               |
| Creativity                                                                                      | 3      | 0                | 0                | +1               | +1               |
| <b>Total</b>                                                                                    |        | 0                | 7                | 3                | 8                |

| Requirement                                                                                     | Weight | Design concept 5 | Design concept 6 | Design concept 7 | Design Concept 8 |
|-------------------------------------------------------------------------------------------------|--------|------------------|------------------|------------------|------------------|
| Must be able to individually push an object weighing <u>18lbs</u> <del>± 2lbs.</del>            | 5      | +1               | -1               | 0                | 0                |
| Must push the asteroid against the wall in quadrant IV within 40 seconds <del>± 5 seconds</del> | 5      | +1               | +1               | +1               | +1               |
| Must be able to collect 3 ± 1 cubes at a time.                                                  | 3      | -1               | -1               | -1               | 0                |
| Manufacturability                                                                               | 3      | -1               | -1               | 0                | 0                |
| Creativity                                                                                      | 3      | +1               | +1               | 0                | +1               |
| <b>Total</b>                                                                                    |        | 7                | -3               | 2                | 8                |

| Requirement                                                                                     | Weight | Design concept 9 | Design concept 10 |
|-------------------------------------------------------------------------------------------------|--------|------------------|-------------------|
| Must be able to individually push an object weighing <u>18lbs</u> <del>± 2lbs.</del>            | 5      | +1               | +1                |
| Must push the asteroid against the wall in quadrant IV within 40 seconds <del>± 5 seconds</del> | 5      | 0                | -1                |
| Must be able to collect 3 ± 1 cubes at a time.                                                  | 3      | +1               | 0                 |
| Manufacturability                                                                               | 3      | +1               | -1                |
| Creativity                                                                                      | 3      | +1               | +1                |
| <b>Total</b>                                                                                    |        | 14               | 0                 |



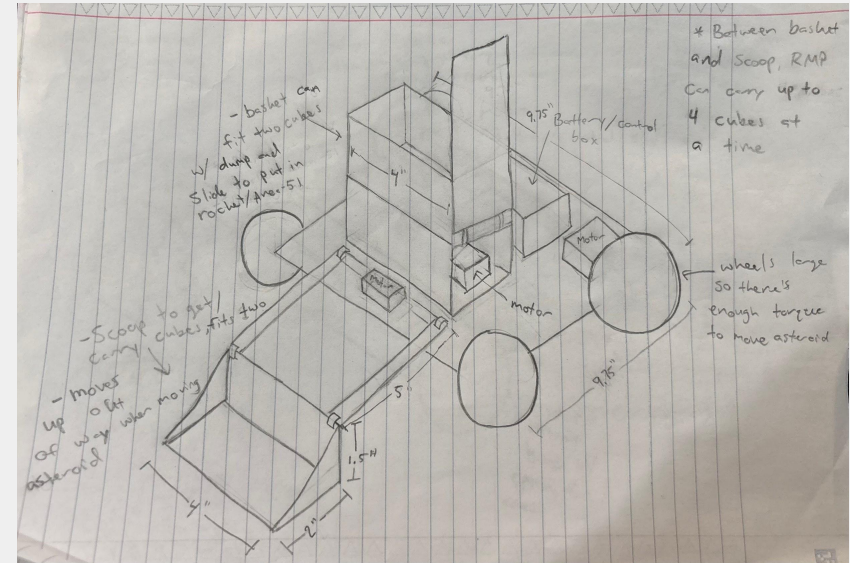
# Functional Requirement Weighting

- Must be able to individually push an object weighing  $18 \text{ lbs} \pm 2 \text{ lbs}$ 
  - Weighted as a 5 because we must be able to push the asteroid as it is essential for the rest of the squad's success to drop the flag and pick up some cubes.
- Must push the asteroid against the wall in quadrant IV within  $40 \text{ seconds} \pm 5 \text{ seconds}$ 
  - Weighted as a 5 because the asteroid must be moved in a timely manner in order for the rest of the squad to complete their tasks in the 5-minute limit.
- Must be able to collect  $3 \pm 1$  cubes at a time
  - Weighted as a 3 because collecting cubes is one of our RMPs main tasks in the squad strategy and it is where we would be getting the most points from.
- Manufacturability
  - Weighted at 3 due to its mixture of being important, while operating under the presumption that none of our designs are wildly outlandish nor impossible to manufacture. In other words, we want a design that can easily be fabricated and we believe that most of our design concepts do not stress the bandwidth of our team's time/energy. This was used as a tie-breaker if two RMP concepts had the same result in the Pugh Chart.
- Creativity
  - We weighted this a 3 because we want our RMP to be unique and not a cookie-cutter solution.



# Pugh Chart Results

- Modification of Design Concept 9
- Score of 14
  - Satisfies all 3 functional requirements
  - Easily manufacturable
    - Geometry is mostly rectangular
- Means of satisfying FRs. are not overwhelming
  - Simple scoop mechanism
  - Multiple motors for wheels
- Adjustments
  - Replaced two wheels with ball caster
  - Redesigned “basket” in the rear
  - Placed control box under basket
  - Implemented ramp system to move cubes from basket to Area 51/rocket



# Decision Variables

- Scoop arm gear ratio
  - First Principles Test
  - Knowing the gear ratio is important to make sure our motor will generate enough torque to lift the arm with 2 cubes in the scoop.
- Drivetrain gear ratio
  - First Principles Test
  - Knowing the drive train gear ratio is important to tuning our RMP to be able to push an object of weight  $18 \pm 2$  lbs within  $40 \pm 2$  sec.
- Depth of Cube Basket
  - Empirical Test
  - Knowing the depth of the basket while taking into account the bounce of the cubes is important to our team achieving our functional requirement of collecting  $3 \pm 1$  cubes.

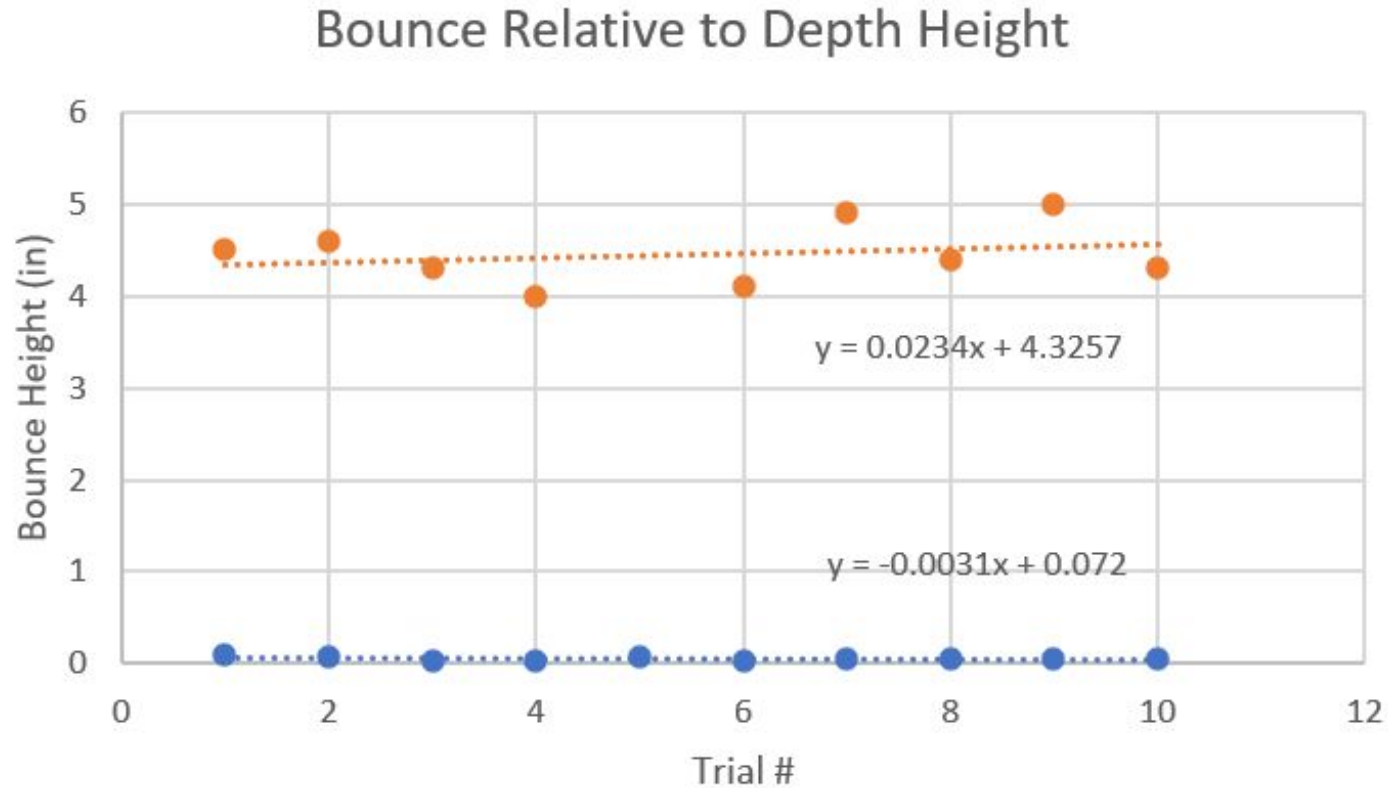


# Empirical Test: Depth of Cube Basket

- Q: Is there a significant bounce from the cube we will have to take into account when selecting the height of basket side?
- This question serves to ensure we meet our functional requirement of collecting  $3 \pm 1$  cubes at a time.
  - Collecting refers to both acquiring the cubes and subsequently housing them
  - If a cube is dropped from the scoop mechanism, there is a possibility that a cube might bounce out, thus preventing us from meeting this functional requirement
- Create a prototype basket out of cardboard
- Made to simulate early-design dimensions of basket
- Consistent drop height of 9" above the base of underlying surface (i.e table)
- Adjusted the depth of basket's bottom surface



# Empirical Test



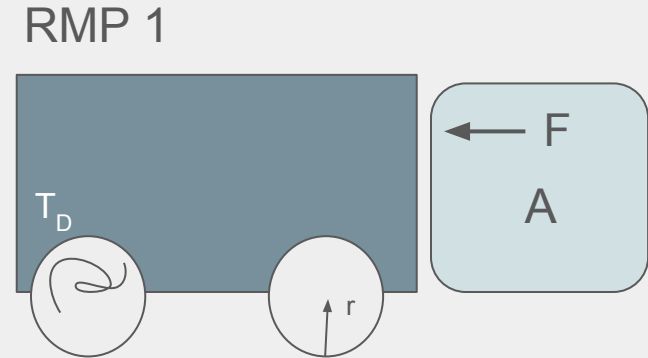
# Conclusions from Empirical Test

- Raising the depth of scoop results in marginal increase of bounce height
  - Less than 1:1 ratio of raising depth vs increased bounce height
- Experiment conducted using cardboard, so the tested material became a concern
  - Q: Will using the Delrin plate differ significantly from prototype basket with regards to dampening the kinetic motion of the cube?
- We found that the Delrin dampened the motion of the dropped cube even more than the cardboard
- We expect that the dimensions of the basket can be selected rather arbitrarily due the expectation that any bounce height will be minimal



# First Principles Test 1: Drivetrain Gear Ratio Analysis

- We need to figure out which gear ratio to use for our RMP to push the asteroid
  - If we have the wrong gear ratio, the asteroid won't move which will not allow us to meet our functional requirement of pushing a 18 lb asteroid.
- Assumptions:
  - $\gamma \approx 0.8$
  - Max efficiency of 0.3
  - Factor of safety of 1.75
  - Force to push asteroid collected with a force probe



# First Principles Test 1: Drivetrain Gear Ratio Analysis

To determine the appropriate gear ratio for the drivetrain, we follow a structured approach:

- Compute the Driving Torque required to move the asteroid.
- Determine the torque at the second stage of the system.
- Compute the required gear ratio.
- Verify if the chosen gear ratio satisfies the torque requirements.

**Step 1: Compute the Driving Torque** The driving torque is calculated using the force applied and the lever arm:

$$T_D = f_s \cdot F \cdot r$$
$$T_D = (1.75)(0.0365125m)(12N) = 0.7668 \text{ Nm}$$

**Step 2: Compute the Torque at the Second Stage**

$$T_{S2} = \frac{V_2}{V_1} \cdot T_{S1}$$

Given:

$$V_{S1} = 6V, \quad V_{S2} = 9V, \quad T_{S1} = 0.0062 \text{ Nm}$$

The torque at stage 2:

$$T_{S2} = 0.0093 \text{ Nm}$$



### Step 3: Compute the Required Gear Ratio

$$T_r = (0.8)(0.3)(0.0093 \text{ Nm}) = 0.002232 \text{ Nm}$$

$$M_r = \frac{T_D}{T_r} = \frac{0.7668}{0.002232} \approx 344$$

### Step 4: Validate Torque Requirement

$$T_D \leq \gamma M T_s$$

$$0.7668 \leq (0.8)(344)(0.0093) = 2.56 \text{ Nm}$$

Since the condition is satisfied, the gear ratio of 344 is appropriate.

**Conclusion:** The computed gear ratio of 344 meets the requirement for moving the asteroid efficiently.



# First Principles Test 2: Arm and Scoop Gear Ratio

- We need to figure out the gear ratio to use with the planetary motor in order for our scoop to lift 2 cubes at a time.
- Using the stall torque of the motor and the mass+center of mass of our arm and scoop scoop we will find the required gear ratio.
- Assumptions:
  - The mass and centre of mass calculated by solidworks is correct
  - The arm mass = 0.3 lbs and scoop mass = 0.04 lbs
  - The arm + scoop's center of mass = 3.091 in from shaft
  - $\gamma \approx 0.8$
  - Safety factor = 1.75
  - Maximum efficiency = 0.2



# First Principles Test 2: Arm and Scoop Gear Ratio Analysis

## Step 1: Compute Lifting Torque

$$T_L = f_s Fr$$

$$T_L = (1.75)(0.34 \text{ lbs} * 3.091 \text{ in} + 0.07 \text{ lbs} * 6.5 \text{ in})(16 \text{ oz}/1 \text{ lb}) = 42.167 \text{ oz-in}$$

## Step 2: Compute Torque at Second Stage

$$T_{s2} = \frac{V_2}{V_1} * T_{s1}$$

Given:

$$V_1 = 3 \text{ V} \quad V_2 = 4.5 \text{ V} \quad T_{s1} = 1.2 \text{ oz-in}$$

Torque at stage 2:

$$T_{s2} = 1.8 \text{ oz-in}$$



### Step 3: Compute Required Gear Ratio

$$T_R = 0.2 * \gamma * T_{S2}$$

$$T_R = 0.288 \text{ oz-in}$$

$$M_R = \frac{T_L}{T_R} = \frac{42.167 \text{ oz-in}}{0.288 \text{ oz-in}} \approx 146.411$$

### Step 4: Validate Torque Requirement

$$T_L \leq \gamma * M * T_{S2}$$

Select a given gear ratio close to  $M_R$

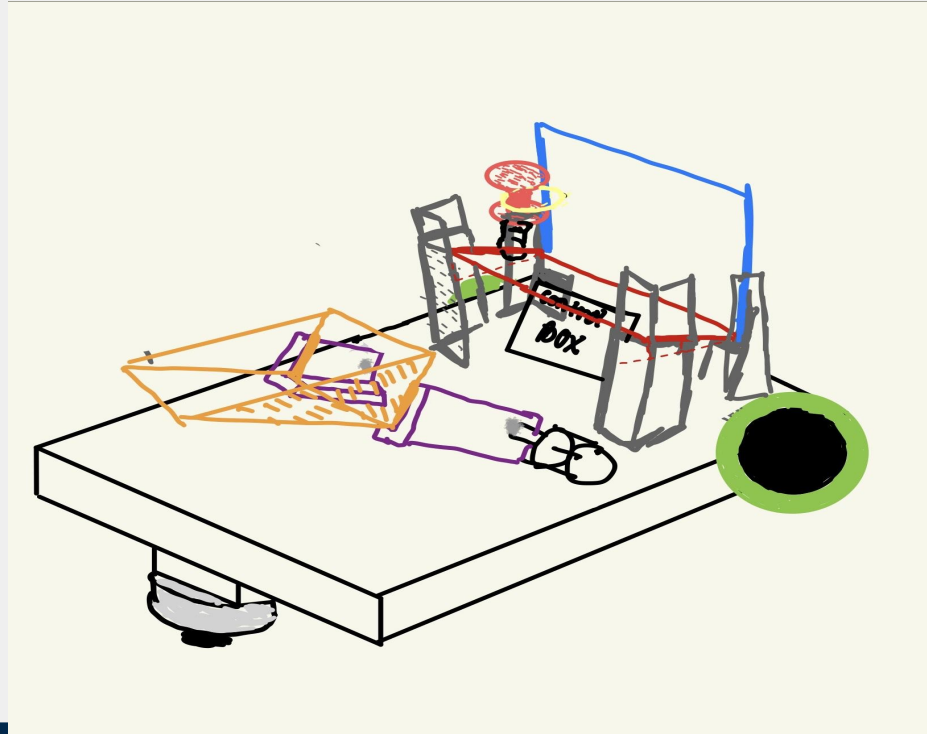
$$M = 100$$

$$42.167 \text{ oz-in} \leq (0.8)(100)(1.8 \text{ oz-in}) = 144 \text{ oz-in}$$

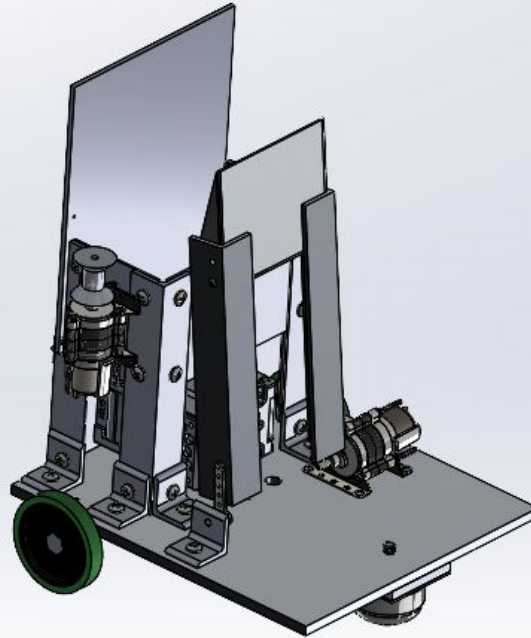
Since the equation is satisfied, a gear ratio of 100 will be enough to lift the arm and scoop with 2 cubes.



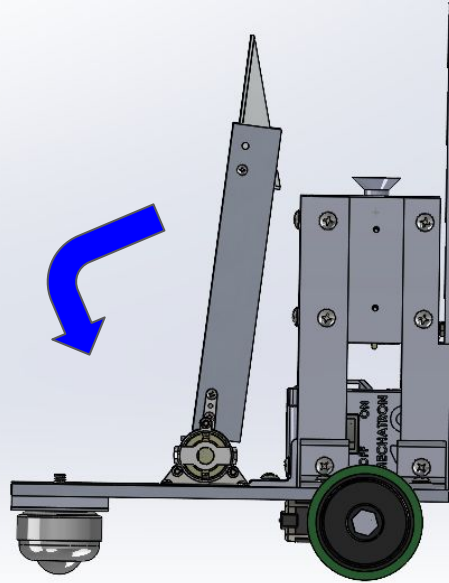
# Final RMP Design



# RMP CAD



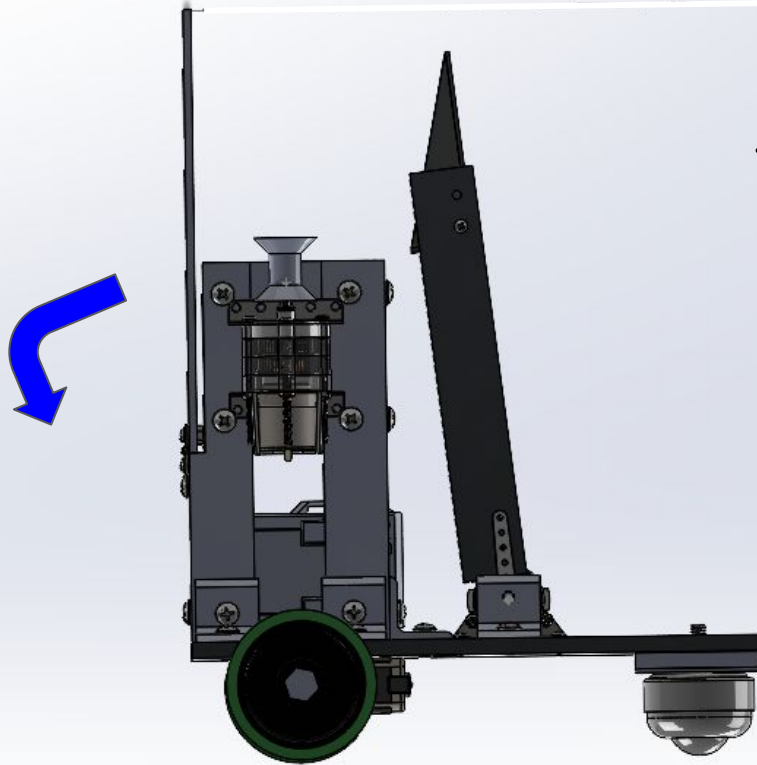
# RMP CAD



The scoop  
rotates up  
and down  
using the  
motor



# RMP CAD

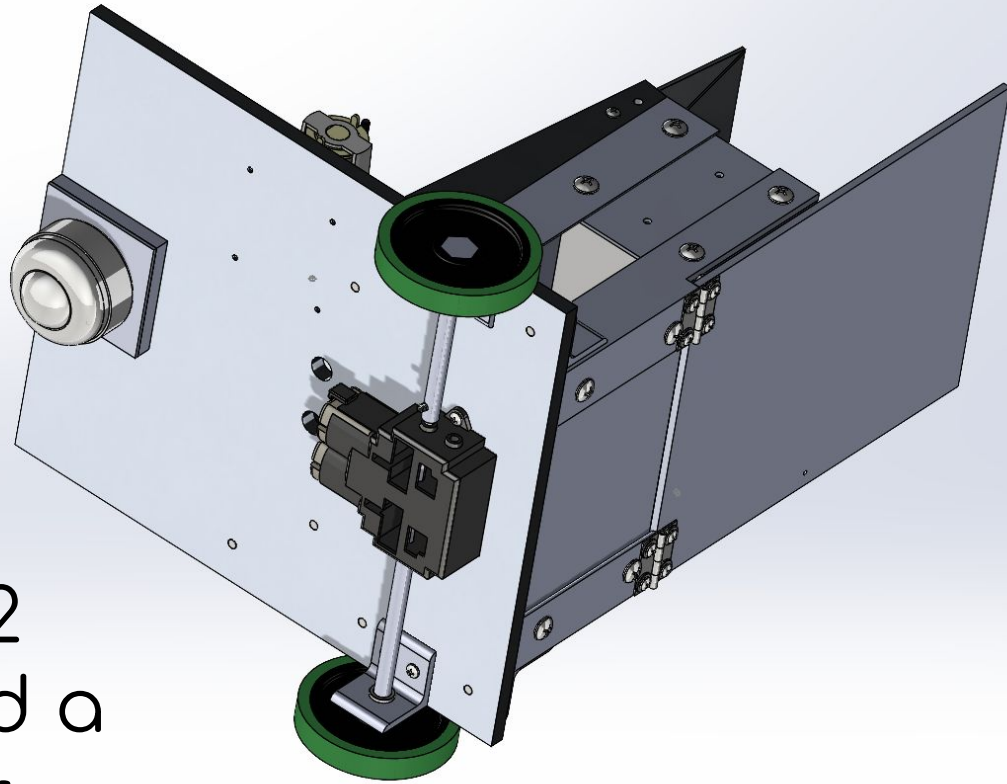


The back slide  
opens like a  
drawbridge

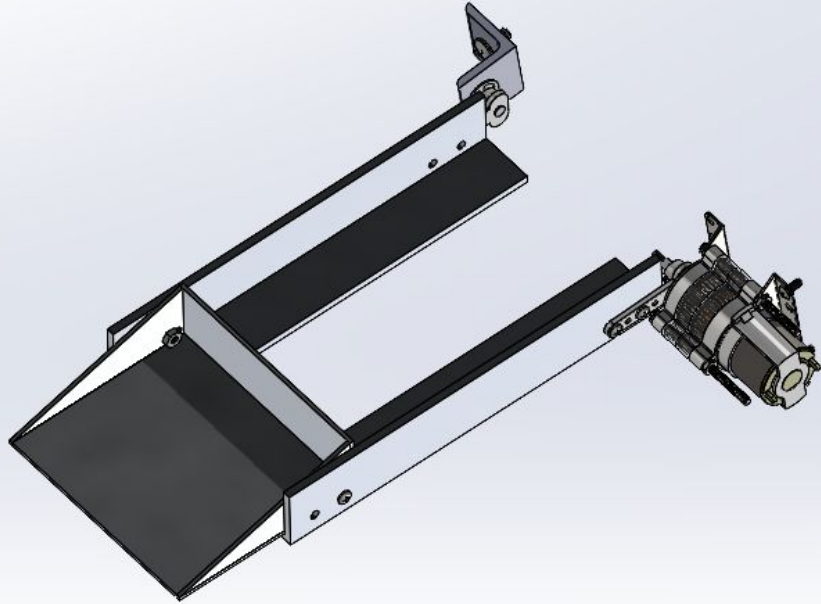


# RMP CAD

The RMP  
drives on 2  
wheels and a  
ball caster



# RMP CAD



The scoop  
is our most  
critical  
subsystem  
because of  
its  
versatility



# Concerns/Challenges

- The biggest concern is our spool system
- The ramp also needs to be stable for the cubes to slide down



# Questions?

